

Prompt Engineering and Large Language Models

Final Examination

Whole course, weighted towards Weeks 6–13 · 25 questions × 4 points = 100 points ·
60% of the course grade

Name: _____ Student No: _____ Date: _____

Instructions. This paper contains 25 multiple-choice questions. Select the **single best** answer for each. Each question is worth equal marks. Time allowed: 60 minutes. No notes or electronic devices.

1. Why is hallucination not simply a bug awaiting a fix?
 - A) The models are still too small
 - B) There is no step in the operation at which truth is checked; plausibility is what is optimised
 - C) Training data has not yet been cleaned
 - D) Users write insufficiently precise prompts
2. In which situation is hallucination **most** likely?
 - A) A widely documented topic in English
 - B) A specific citation on a thinly covered topic
 - C) A request for a summary of provided text
 - D) A request to rewrite a paragraph more simply
3. A question is phrased “Explain why X causes Y”, when in fact X does not cause Y. What is the likely result?
 - A) The model corrects the premise
 - B) The model produces an explanation, accepting the premise
 - C) The model refuses to answer
 - D) The model reports low confidence
4. Why does the same content cost more tokens in Turkish than in English?
 - A) Turkish words are longer on average
 - B) Agglutinative stacking produces assembled words that appeared rarely in training
 - C) Turkish requires a separate tokenizer pass
 - D) Turkish uses more punctuation
5. Which is **not** held in the context window?
 - A) The conversation so far
 - B) Attached documents
 - C) Hidden instructions added by the product
 - D) The model’s training data
6. Why is repeating a question and receiving a similar answer weak evidence of correctness?

- A) The model remembers its previous answer
 - B) Consistency measures how strongly a pattern is represented, not whether it is true
 - C) The second answer is always shorter
 - D) The context window has changed
7. Adding “be accurate and use only reliable sources” to a prompt about a thinly covered topic will most likely:
- A) Improve factual accuracy
 - B) Produce a refusal
 - C) Make the answer sound more authoritative without improving it
 - D) Trigger a web search
8. Which is a well-formed negative constraint?
- A) Do not be vague
 - B) Do not write anything unprofessional
 - C) Do not cite any case or regulation by name
 - D) Do not disappoint the reader
9. An answer about Turkish procedure accurately describes the procedure of another country. This is:
- A) Fabrication
 - B) Displacement
 - C) Omission
 - D) Silent failure
10. A legal memo contains a case name in correct citation format for a case that was never decided. This is:
- A) Fabrication
 - B) Displacement
 - C) Over-reading
 - D) Indirect injection
11. Which failure is hardest to detect outside your own area of expertise, and why?
- A) Fabrication — because citations look real
 - B) Displacement — because foreign systems are unfamiliar
 - C) Omission — because there is no incorrect statement to identify
 - D) Overconfidence — because tone is uniform
12. A grounded assistant answers a question your documents do not cover, producing a confident response with no quotation. This is:
- A) Retrieved the wrong passage
 - B) Over-read the passage
 - C) Answered from memory
 - D) Silent failure
13. Your source says “the committee *may* require additional evidence”. The assistant reports “the committee *requires* additional evidence”. This is:

- A) Fabrication
 - B) Over-reading the passage
 - C) Answered from memory
 - D) Displacement
14. A three-step workflow receives faulty input at step one and produces a complete, confident, well-formatted output. Nothing reports an error. This is:
- A) Wrong stopping point
 - B) Silent failure
 - C) Context overflow
 - D) Direct injection
15. What do all nine failure modes covered in this course have in common?
- A) They occur only in long conversations
 - B) They are invisible in the output
 - C) They can be prevented by better prompting
 - D) They occur only in grounded systems
16. Which check would most reliably expose a poisoned document summary?
- A) Asking the model whether it was manipulated
 - B) Regenerating the summary several times
 - C) Checking each claim against a quoted passage in the source
 - D) Reducing the temperature
17. What is the most significant thing lost when a chatbot is replaced by an agent?
- A) Response speed
 - B) The unconscious checking you performed on every intermediate output
 - C) The ability to attach documents
 - D) Access to the context window
18. If each step is 95% reliable, approximately what is end-to-end reliability over ten steps?
- A) 95%
 - B) 80%
 - C) 60%
 - D) 40%
19. Which control does **not** rely on the system behaving as intended?
- A) Instructing it to be careful
 - B) Asking it to verify its own output
 - C) Connecting only the narrowest tools required
 - D) Requesting quotations for each claim
20. Where should a human be placed in an AI workflow?
- A) At the end, reviewing the output
 - B) Before the irreversible action

- C) At the start, writing the prompt
 - D) Wherever capacity allows
- 21.** Which is **not** a criterion for a good first automation candidate?
- A) The task is genuinely repetitive
 - B) The answers exist in documents you can point at
 - C) A wrong answer is recoverable
 - D) The task is the most intellectually difficult in your field
- 22.** In a well-designed AI workflow, most decisions concern:
- A) Which model to use
 - B) What the system must not do
 - C) How to phrase the prompt
 - D) Which tool provider to select
- 23.** Why can a Domain Pack increase risk as well as usefulness?
- A) It consumes context window space
 - B) Professional formatting makes wrong content more persuasive
 - C) It slows the system down
 - D) It prevents the system from refusing
- 24.** Why is indirect injection more serious for an ordinary user than direct injection?
- A) It is technically harder to perform
 - B) The malicious instruction is hidden in content the user never sees, so there is no point at which they could refuse
 - C) It cannot be defended against at all
 - D) It only affects grounded systems
- 25.** Under this course's account of academic integrity, the central question is:
- A) Whether AI was used at all
 - B) What the assessment is measuring, and what you are certifying by submitting
 - C) Whether the use can be detected
 - D) How much of the text was generated